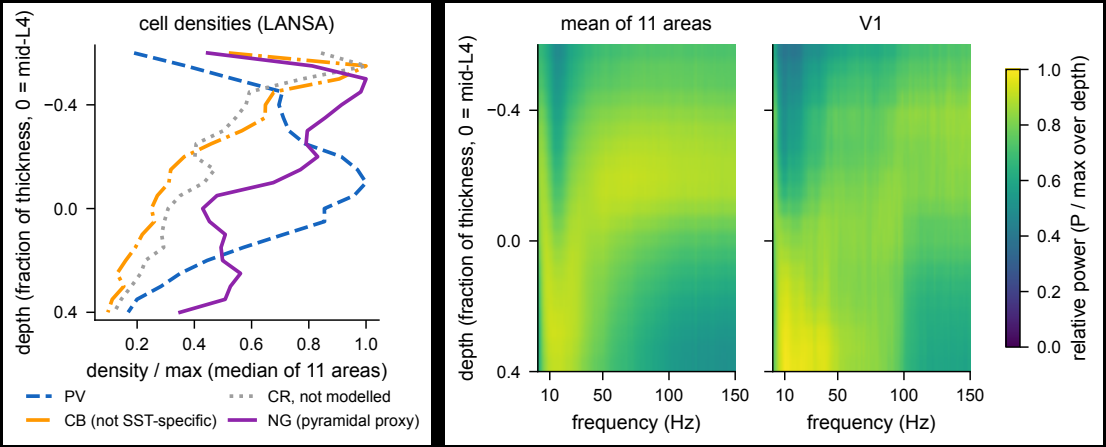
Testing cell-type density as a circuit mechanism of primate spectrolaminar oscillations

A column with density-sized units and an imposed SOM/PV ramp puts alpha-beta in the deep half on the reference seeds, and leaves gamma near layer 4 weak. A three-source readout could carry the observed gamma profile; readout depth remains a confound.

Hamed Nejat
Bastos Lab · Vanderbilt University · November 10, 2026

Preliminary, unpublished results.

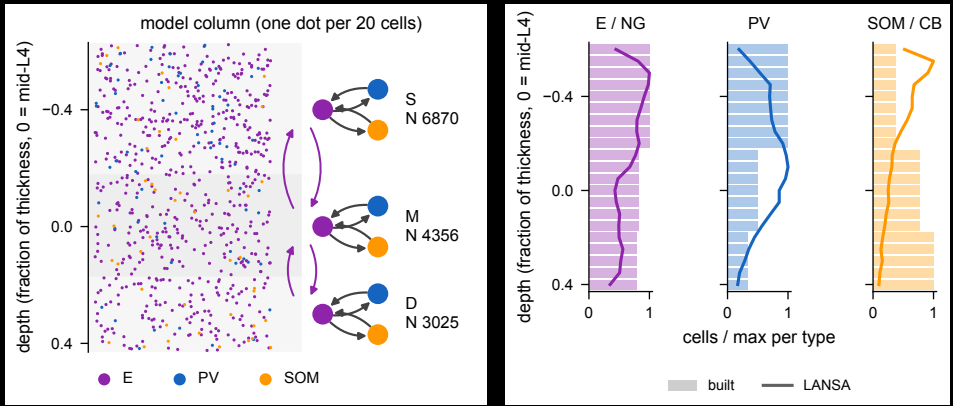
Macaque cortex: gamma is relatively strongest above L4, alpha-beta below it



Left: cell densities by depth (median of 11 areas). Right: LFP power / its maximum over depth.

LANSA data, Mulvey et al. 2024; motif after Mendoza-Halliday et al. 2024.

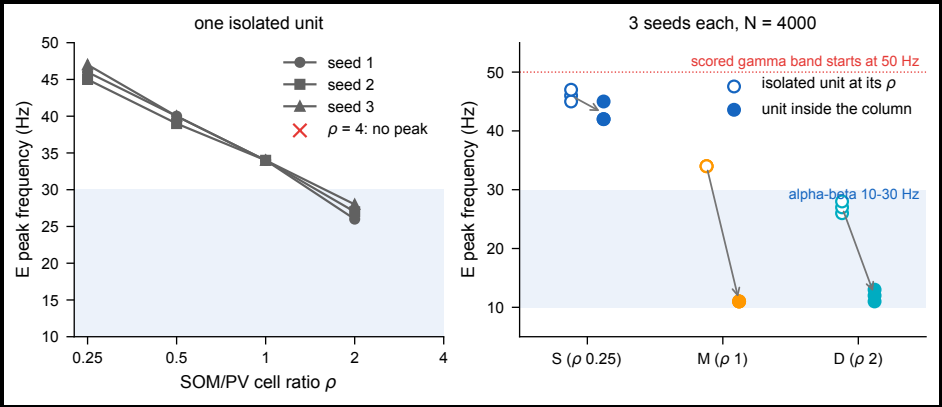
Unit sizes come from these densities; the SOM/PV ramp is imposed



4000 E cells per unit on average; 14 251 cells (S 6870, M 4356, D 3025). SOM/PV 0.25, 1, 2: imposed.

AdEx units at −0.4, 0, +0.3; in-degree scales with density by an assumed rule. E/PV/SOM: S 5785/868/217, M 3668/344/344, D 2547/159/319. Built PV follows unit size × ρ; SOM follows the ramp; CB is calbindin, which labels more than SST cells. NG: pyramidal proxy.

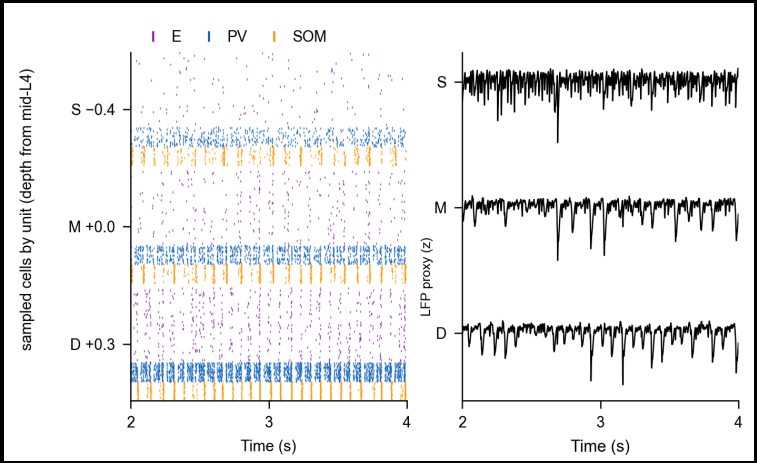
Isolated, SOM/PV lowers the rhythm; in the column, M and D peak in alpha-beta



At the imposed ratios, M and D peak at 11–12 Hz in the column, S at 43 Hz (below the scored band); cause inferred: coupling and density scaling.

Matches its reference (34.0 vs 33.8 $\pm$ 1 Hz); -6.27 Hz/octave for ρ 0.25–2; $\rho = 4$ silent ($\times$): inconclusive as specified.

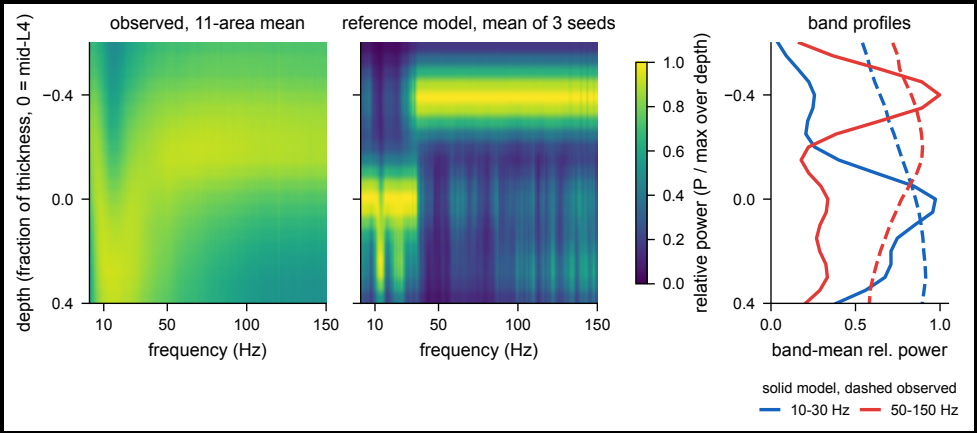
The column fires sparsely: E cells below 1.6 Hz, interneurons faster



Rates over 3 seeds: E 0.39–1.58 Hz, PV 5.6–22.6 Hz, SOM 4.7–9.4 Hz (lowest at S, highest at D).

Reference column, seed 1; 2 s of the 8 s run shown (first 2 s discarded). Left: sampled cells; right: per-unit LFP proxy (weighted synaptic currents).

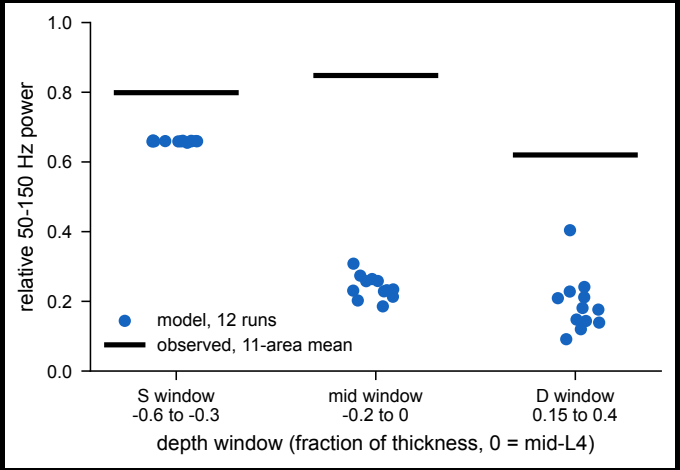
The column puts alpha-beta in the deep half and most gamma at S



r_{ab} 0.83–0.86 on seeds 1–3; flat density gives 0.84 as well.

r_{ab}, r_{gm}: depth correlation of the 10–30 / 50–150 Hz profile with the observed one. Model maps: LFP proxies read through the three-source Gaussian kernel, σ = 0.15. Seed means, 3 seeds; stationary Poisson drive, whereas the observed maps are peri-stimulus. The reference was not rerun on fresh seeds; the later selected setting lost r_{ab} there (0.35–0.47).

Near layer 4, gamma is weak: 0.19–0.31 in 12 runs against 0.848 observed



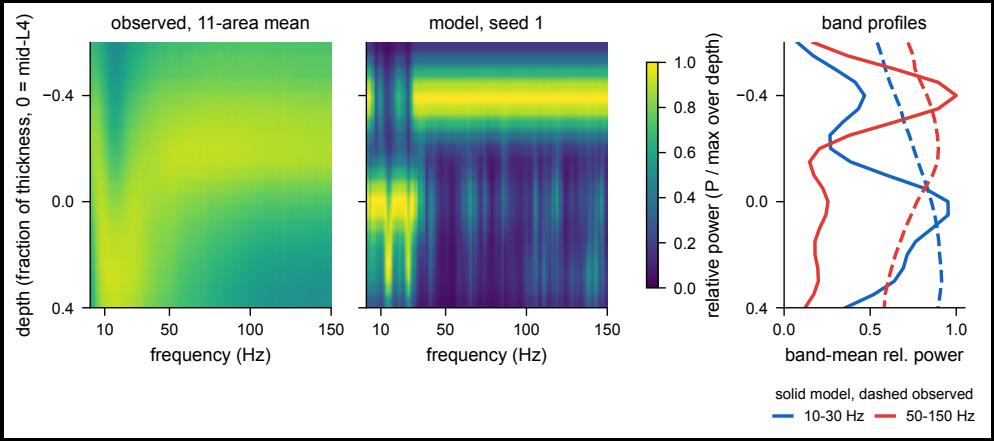
Cross terms carry 0–12% of this power; M supplies 72–81%.

S holds the maximum at 97–101 of 101 gamma frequencies.

M's gamma is small relative to S's.

9 size-ladder runs (N 1000/2000/4000 E per unit $\times$ 3 seeds) and 3 confirmation runs; 50–150 Hz, depths -0.2 to 0.

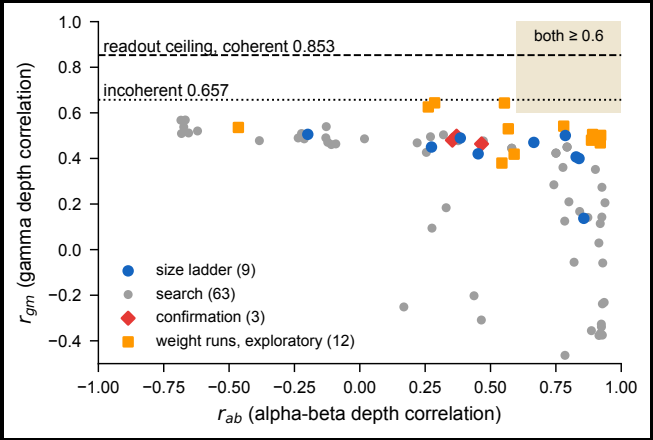
Similarity is scored by one rule, fixed before the search, on the whole map



Pass = crossing $>$ 2 SE, r_{ab}, $r_{gm} \geq 0.6$, true unit order first, map r above null and control.

Crossing: the alpha-beta minus gamma profile is below -2 SE at the top contact and above +2 SE at the bottom. Also: both halves agree. Null: random smooth unit spectra (95th percentile); control: every unit given S's spectrum. Shown: B = 1.41, SOM→E 15 ms in all units, seed 1: map r 0.36, r_{ab} 0.75, r_{gm} 0.42; fails.

No run passes; $r_{gm} \geq 0.6$ (σ 0.15) only in 3 weight runs, none with the crossing



87 scores, 78 distinct runs: size ladder 9, search 63 (54 new runs), confirmation 3, weight runs 12 (13 settings with their start, page 11). The 12 control runs are on pp. 12–13.

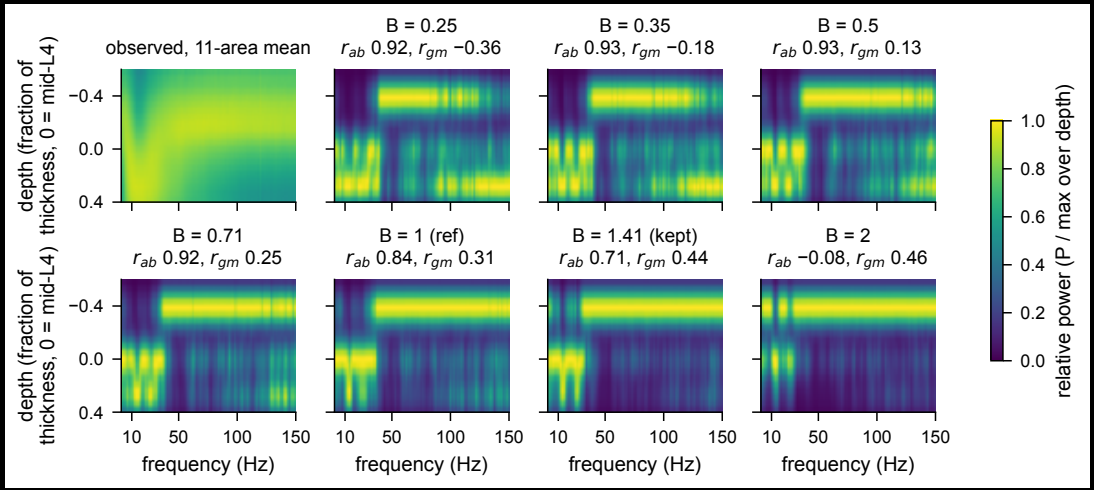
Largest r_{gm} outside the weight runs: 0.57.

Whole-map ceiling, map r (r_{proto}): 0.854 coherent, 0.693 independent.

Gamma-profile ceiling of the readout: r_{gm} 0.853 coherent, 0.657 independent (σ = 0.15).

Ceiling = the best any source covariance reaches, from the observed data alone. At σ 0.20 / 0.25, $r_{gm} \geq 0.6$ in 9 / 11 of the 12 ladder and confirmation runs (none reaches the threshold at 0.15); weak mid-depth gamma holds at every width.

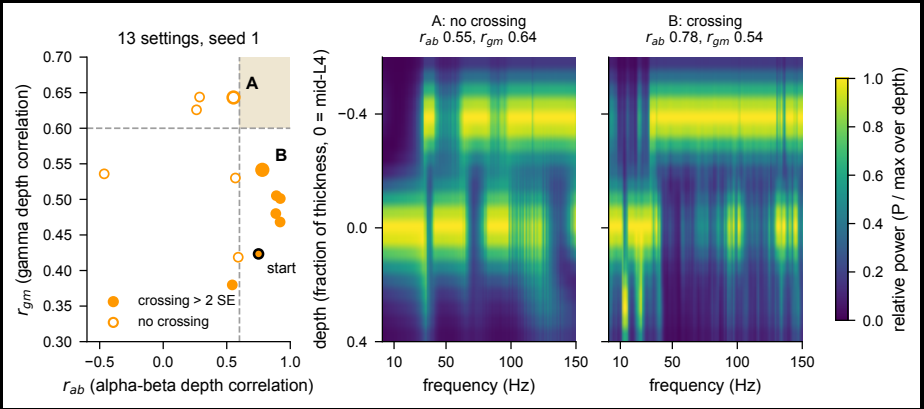
Tuning local in-degree trades alpha-beta against gamma; no level passes



In-degree $\propto$ density^{B}; raising B trades r_{ab} for r_{gm}. B 1.41 kept.

7 levels $\times$ 3 seeds, N = 4000, 8 s. Final selection (B 1.41, SOM$\rightarrow$E 21 ms in D) on fresh seeds: r_{ab} 0.35–0.47, 0/3 pass.

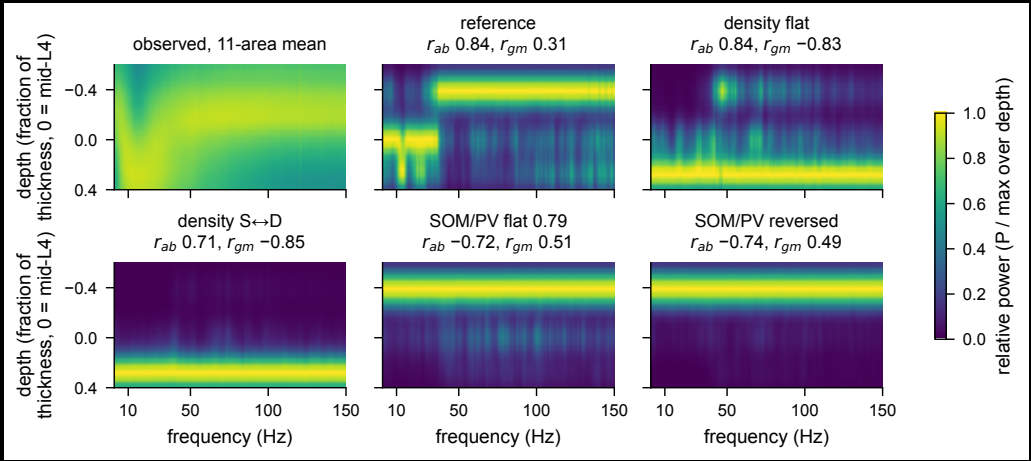
Weight changes (mainly at M): $r_{gm} \geq 0.6$ only without the crossing



$r_{gm} \geq 0.6$ only without the crossing (A); with the crossing kept, r_{gm} 0.42 → at most 0.54 (B; r_{ab} 0.78). All 13 settings fail the scorer.

Exploratory, seed 1, 8 s; base B 1.41, SOM→E 15 ms in all units (page 8). A: M E→E ×1.68, S, M E→SOM ×0.5. B: M E→E ×1.68, S E→SOM ×0.5, M PV→E ×1.25. M's recurrent excitation sets both bands (inferred).

Changing density or the imposed SOM/PV ramp breaks the frequency split

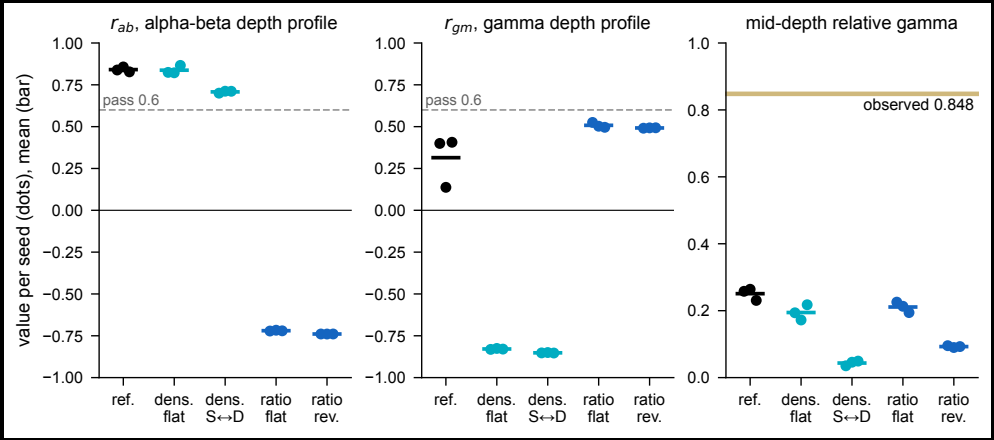


Density arms: D dominates most frequencies; SOM/PV arms: S does (read from the maps).

Seed means, 3 seeds; controls descriptive, uncorrected. One rule changed per control; total cells fixed, unit sizes may change.

Flat ratio 0.79 = the ramp's geometric mean.

Each change hands the map to one unit; none raises gamma near L4



No arm raises gamma near L4 (0.04–0.23; reference 0.23–0.26); flat density keeps r_{ab} through deep dominance (inferred); the ratio arms' r_{gm} near 0.5 comes from S dominating most frequencies.

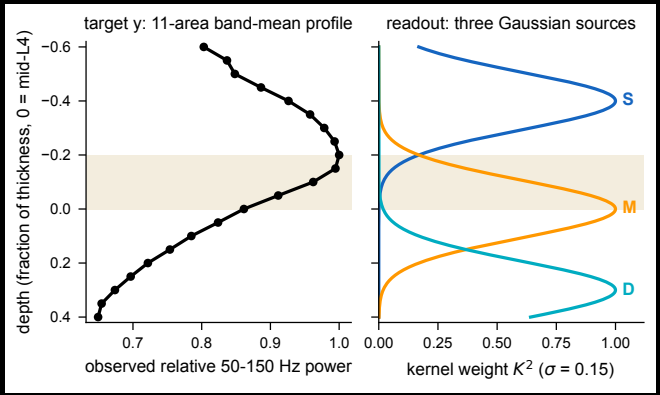
Black: reference; colour: dominant unit (D cyan, S blue). Ratio = imposed ramp; arms also change sizes, in-degrees, rates.

Is the readout the bottleneck?

No circuit setting, control or weight change passed the scorer or reached the observed mid-depth gamma (0.848).

Next, without simulation: could any mix of three depth sources reproduce it? If none can, the readout is the bottleneck at the declared depths. Whether those depths are right is tested last.

Step 1: target = observed gamma profile; readout = three Gaussian sources



$$P_z = k_z^\top C\, k_z, \quad C = LL^\top \succeq 0$$

$$k_{zu} = e^{-(z-z_u)^2/2\sigma^2}$$

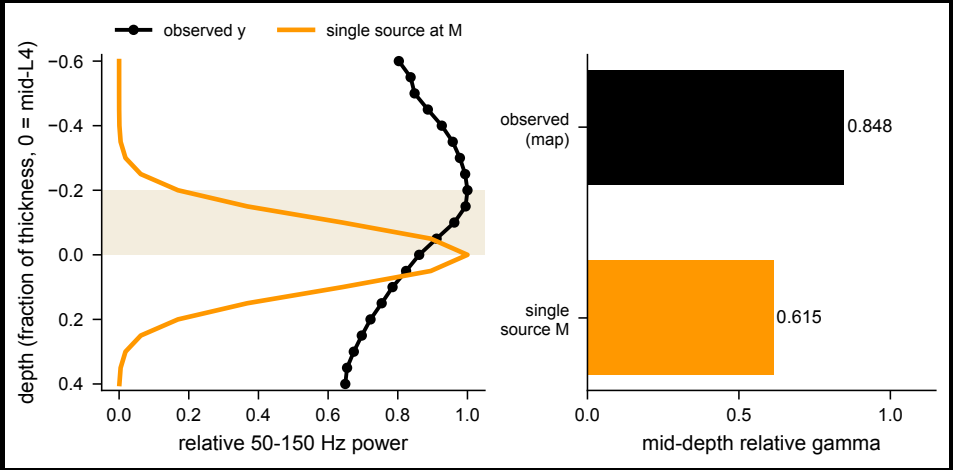
$$R_z = P_z / \max_z P_z$$

z: contact depth (21); $u \in \{S, M, D\}$; C: source covariance, one for 50–150 Hz; mid = mean R_z, $z \in [-0.2, 0]$ (gold band).

No simulation: search C for the best fit this readout can represent.

Target y = observed 50–150 Hz band mean / max (mid 0.946); per-frequency map mid 0.848.

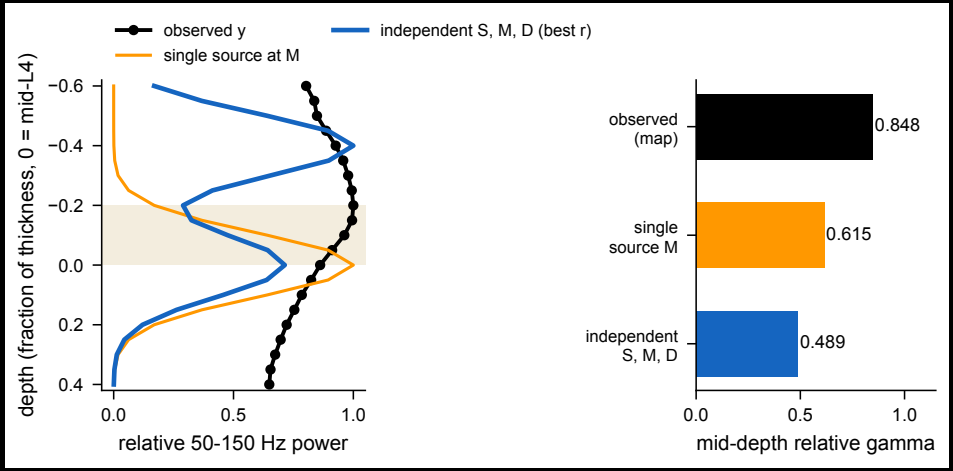
Step 2: a single source at M caps mid-depth gamma at 0.615



σ = 0.15. Raising M's power, even 10^4-fold, tends to this kernel limit, 0.615, below the observed 0.848.

Curve: band-mean profile / its maximum (mid 0.946); bars: per-frequency map (mid 0.848); different normalisations.

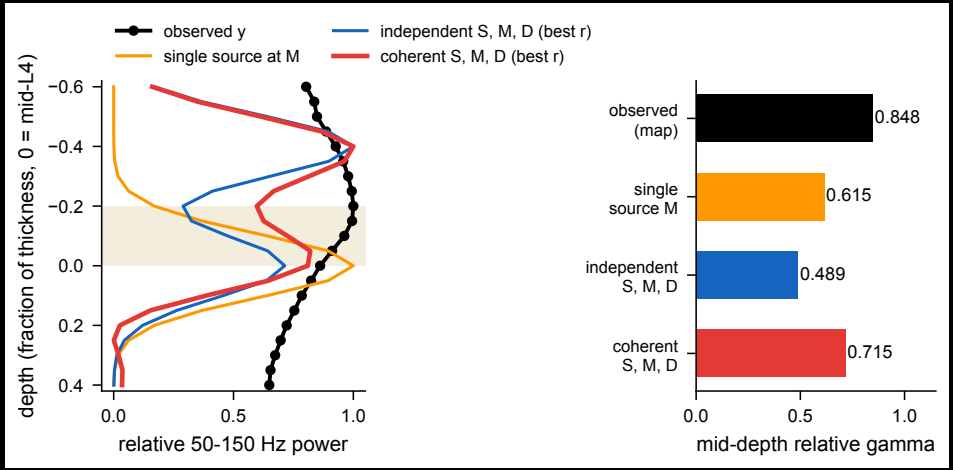
Step 3: independent sources reach $r = 0.657$, with mid-depth gamma 0.49



$\sigma = 0.15$. Best diagonal C: S 58%, M 42%, D 0%; $r_{gm} \geq 0.6$ is passable without coherence.

48 optimizer starts.

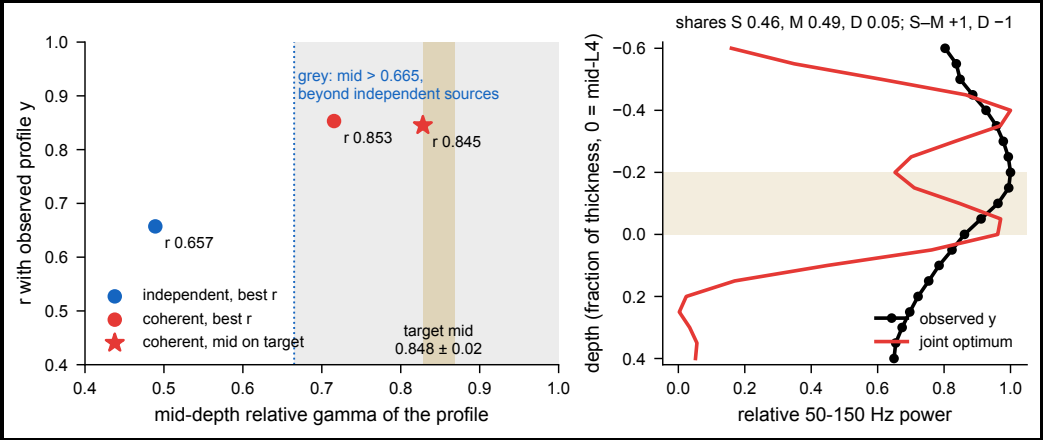
Step 4: coherent sources reach $r = 0.853$; that optimum has mid-depth 0.72



$\sigma = 0.15$. Controls: 5 random profiles recovered at $r \geq 0.99999$; alternating profile r 0.12.

A least-squares fit gives only r 0.45. Optimum rank one: S–M coherence +1, D −1.

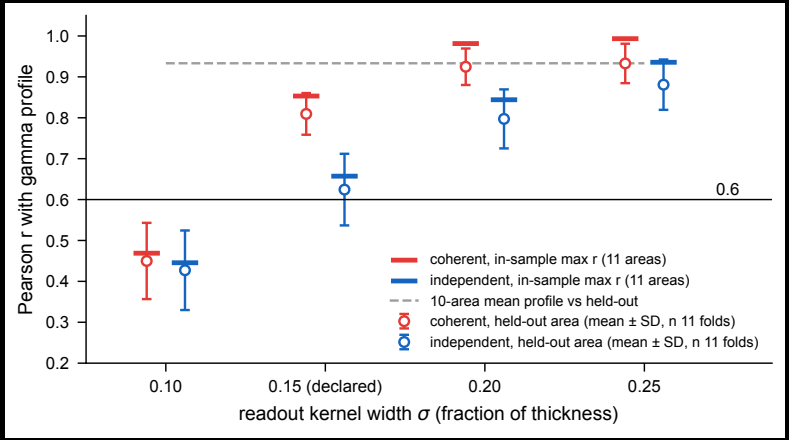
Step 5: at σ = 0.15 the observed mid-depth level needs coherent sources



Best *r* with mid within 0.02 of 0.848: 0.845 at mid 0.828 (edge; coherent). Independent: mid $\leq$ 0.665.

Numerical optimum; maximality unproven; band-mean level 0.946 out of reach; at σ = 0.20 both meet it (mid 0.868 coherent, 0.828 independent).

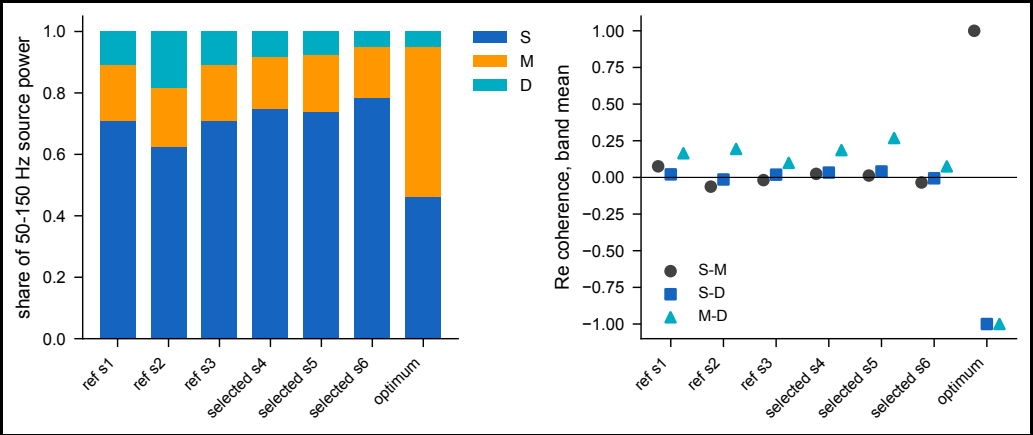
Step 6: held-out areas reach $r \geq 0.6$: coherent 11/11, independent 7/11



σ = 0.15, n = 11 folds, mean $\pm$ SD: coherent 0.81 $\pm$ 0.05, independent 0.62 $\pm$ 0.09.

10-area mean vs held-out: 0.93 $\pm$ 0.06, above the fitted readout's 0.81. Tests the target fit only. Folds share 9 of 10 training areas; SD descriptive.

Step 7: the circuit is far from the readout optimum



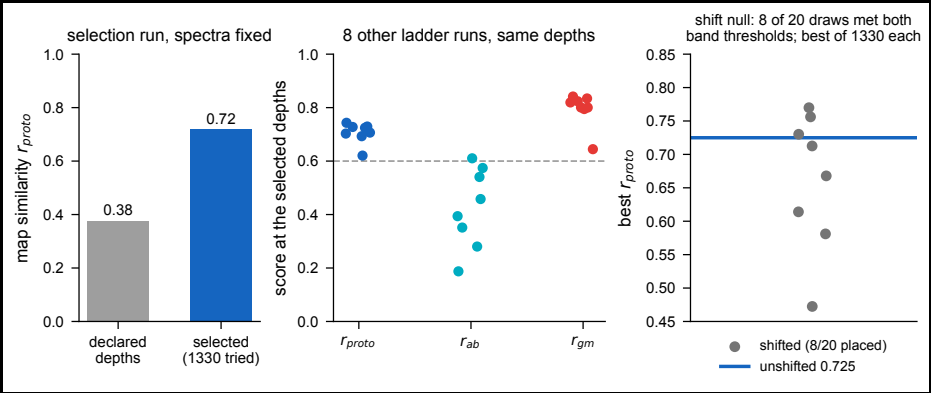
Circuit: S 63–78%, M 17–19% of gamma power, Re coherence -0.06 to 0.27; optimum 46/49/5%, ± 1.

At the declared depths and under this readout, the gap is in the circuit's spectra.

ref s1–s3: reference column, seeds 1–3; selected s4–s6: selected setting, fresh seeds 4–6. 50–150 Hz unit cross-spectra.

Conditional on the three-source readout; its unit-dependent filtering is unvalidated.

Readout depth is a confound: moving it alone nearly doubles map similarity



Spectra fixed: map r 0.38→0.72, indistinguishable from a frequency-shift null ($p \approx 0.19$, 20 draws, best of 1330 each).

Depths S −0.25, M −0.05, D 0.20. Null on the incoherent spectra (unshifted best 0.725, the baseline here): unit spectra circularly shifted, power kept; 12/20 met none, 3 beat 0.725; 20 draws is a weak test. At the selected depths r_{ab} reached 0.6 in 1 of the 8 other runs.

The tested density-informed column does not reproduce the primate motif

- Made: alpha-beta in the deep half on the reference seeds (r_{ab} 0.83–0.86), but flat density does too (0.84). It rests on the imposed SOM/PV ramp, which runs against the measured CB/PV profile; reversing it breaks it. The selected setting lost it on fresh seeds (0.35–0.47); the reference was not rerun.
- Disrupted: changing density or the ramp hands the map to one unit; no band moves alone.
- Missing: gamma near L4, 0.19–0.31 vs 0.848. Weight changes reached $r_{gm} \geq 0.6$ only without the crossing; with it, at most 0.54. Every setting fails the scorer.
- Readout (σ = 0.15, declared depths): coherent sources reach r 0.853 (0.845 with mid near target), independent 0.657; the circuit's Re coherence is -0.06 to 0.27. Moving the depths alone: map r 0.38$\rightarrow$0.72, indistinguishable from a shift null.

Scope and limits

- Tested: one three-unit E/PV/SOM column. Varied: unit sizes, in-degree rule, imposed SOM/PV ramp, SOM→E kinetics, network size. Not tested: PV→E kinetics, E/I balance, PV or SOM silencing, VIP cells.
- Drive: every unit gets the same stationary Poisson drive, while the observed maps come from 1 s peri-stimulus windows with timed laminar inputs; part of the mismatch may reflect drive.
- Target: the LANSA mean; the motif's generality is disputed.
- The readout conclusion is conditional on the three-source readout. This says nothing about where gamma is generated in macaque cortex.
- Next: a middle-layer gamma generator (PV kinetics, rate-matched control) and source geometry.

Sources

Page	Manuscript item; records (Table S6) and data
2–3	Fig. 1; LANSA densities and LFP (Mulvey et al. 2024); builder C1
4	Table 1; isolated-unit check R1, seal S1; SOM/PV sweep R2, seal S2; unit peaks R4
5–7	Fig. 2, Results; run rates R3; size-ladder verdicts R6; mid-depth gamma D1 (code C3); cross terms D2 (C4)
8–10	Fig. 3, Fig. S4; scorer C2; ceilings R5; verdicts R6, R7, R8; search seal S4, selections R3
11	Fig. 5, Table 5; exploratory weight runs R8
12–13	Fig. 6, Table 6; controls seal S7; per-run outcomes D8 (C12); engine check D9; maps R11
15–19	Results, readout capacity; D3 (C5); joint constraint D5 (C7); grid check D7 (C11)
20	Table 8; leave-one-area-out D4 (C6)
21	Table 7; circuit vs optimum D5 (C7)
22	Results, geometry; findings log N1 §7

Manuscript in preparation (draft 2026-11-10). Figures redrawn from these records by the deck's figure script.